Machine Learning Approach for Predicting Stable Distributions on Multiferroic $Bi_6Ti_3Fe_2O_{18}$

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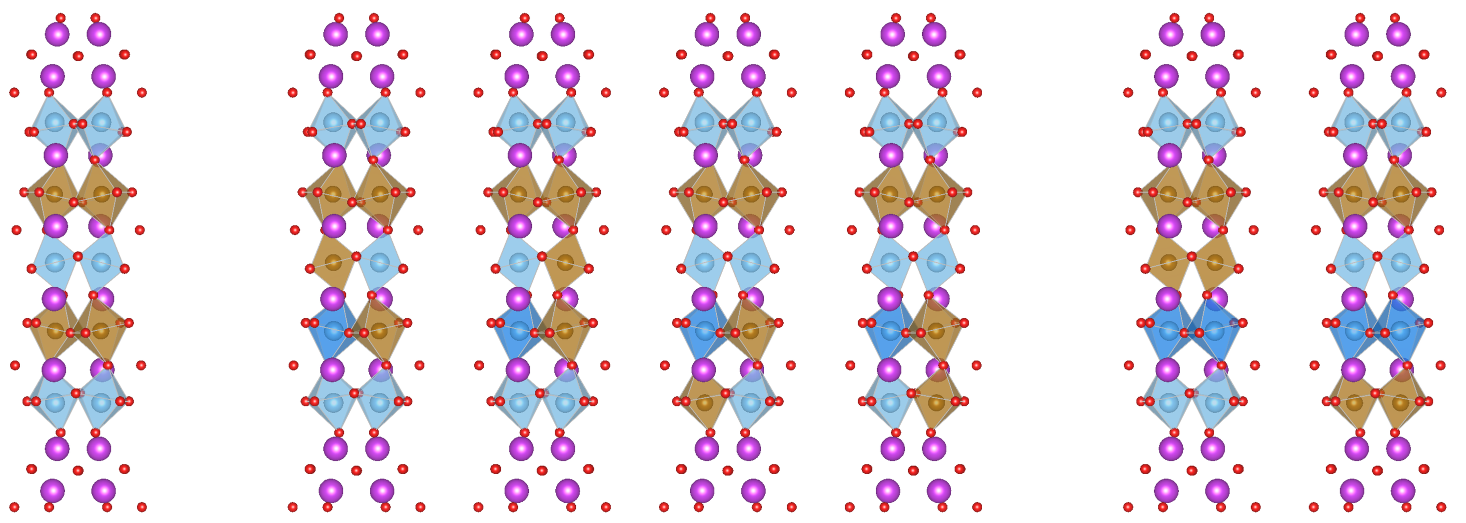
Introduction

In this project, we explore the Aurivillius Phase Multiferroic $Bi_6Ti_3Fe_2O_{18}$ with m=5 perovskite layers. The model uses Density Functional Theory (DFT) for solving the quantum mechanic equations of the material, however this process is time and computationally expensive. We use machine learning models to predict the energy of different configurations in order to overcome the computational challenge.

Structure of $Bi_6Ti_3Fe_2O_{18}$ under different configurations:



config.1 config.11 config.12 config.13 config.14 config.21 config.22

Atoms: •Bismuth •Titanium •Iron •Oxygen

Creating the Dataset

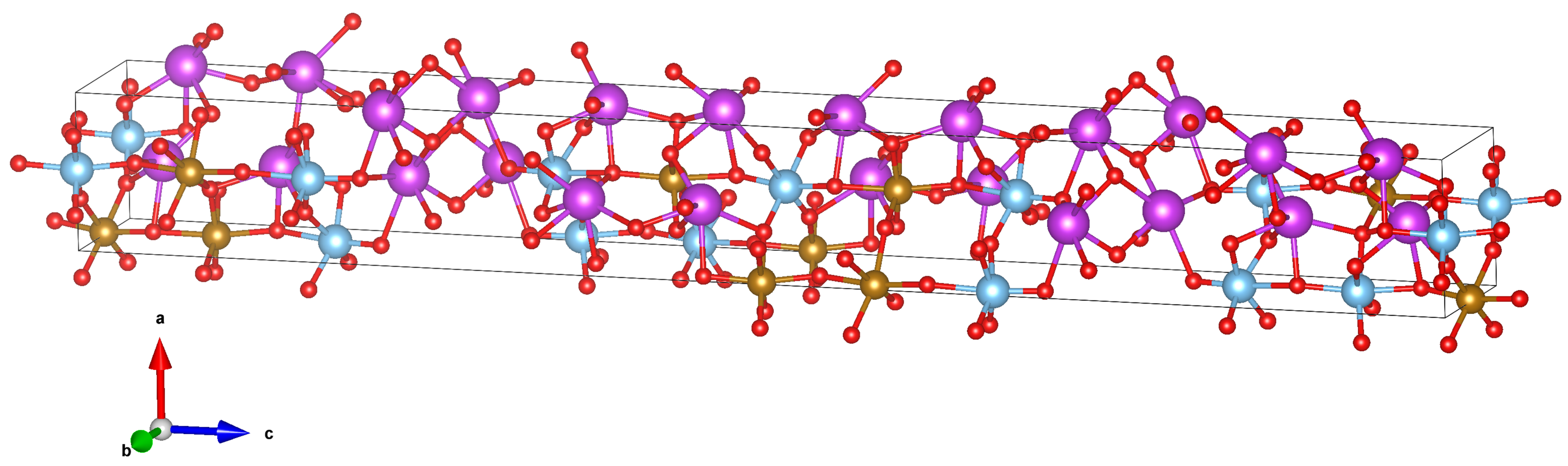
The samples of the dataset are based on unit cell size, the atom's coordinates, and the energies. Each coordinate is multiplied by a weight function regarding properties of each element. The total number of samples is 1680.

Dataset:

Unit Cell **Coordinate Matrix** **Energy**

$$\{a_i,\, b_i,\, c_i\} \oplus \{x_j,\, y_j,\, z_j\} \times W_j \oplus E_k$$

Unit Cell:

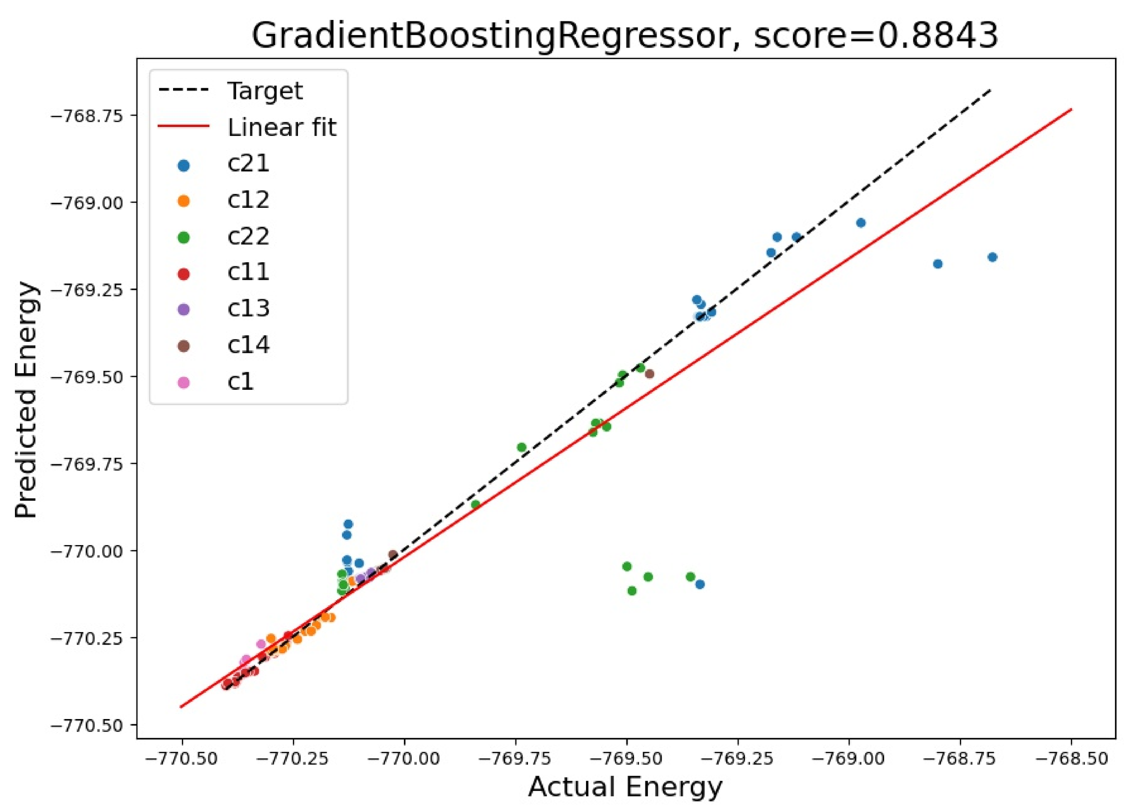


Results

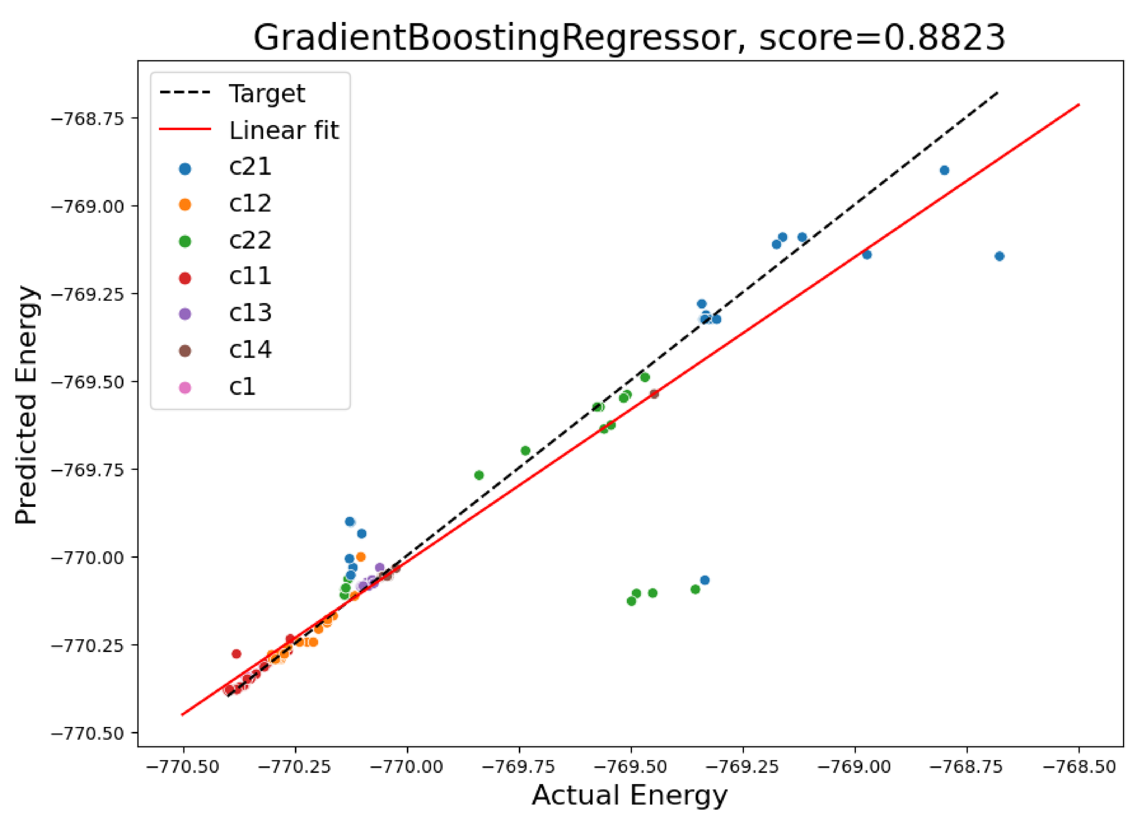
Gradient Boosting Regression:

Gradient Boosting Regression was the most reliable model, achieving a prediction accuracy, measured by R2 validation, of 86.99%.

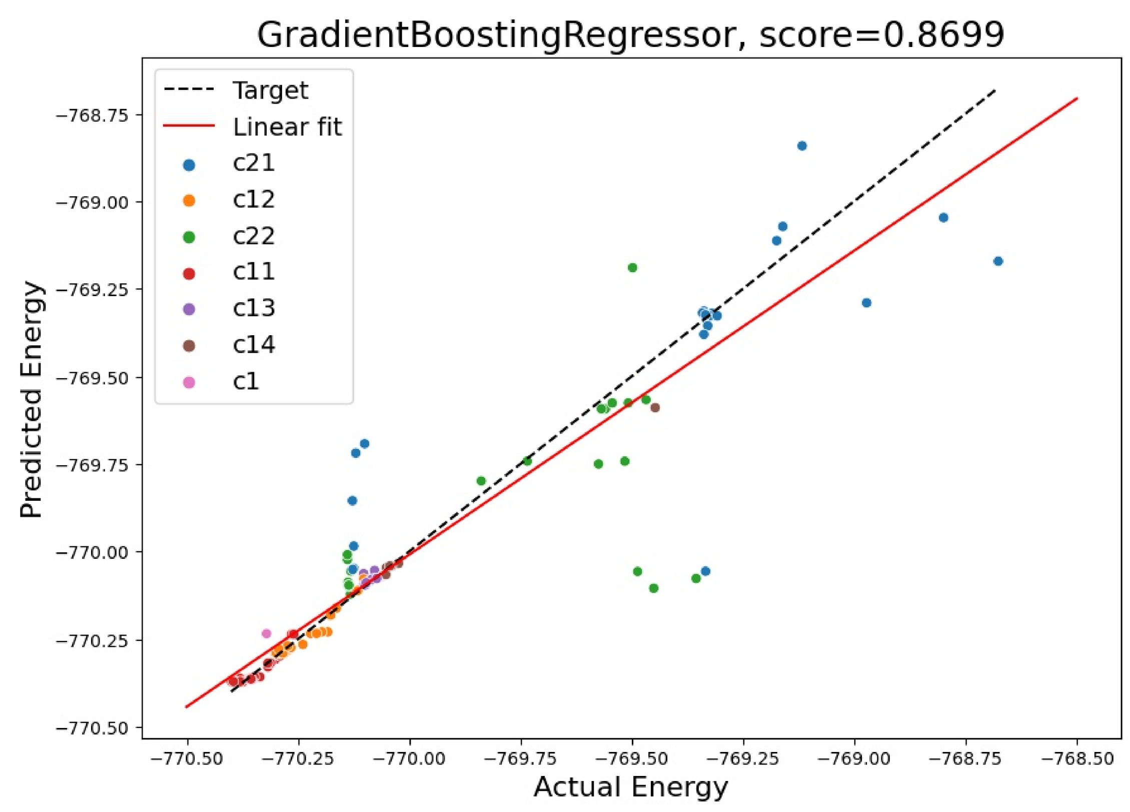
3989 Features



1463 Features

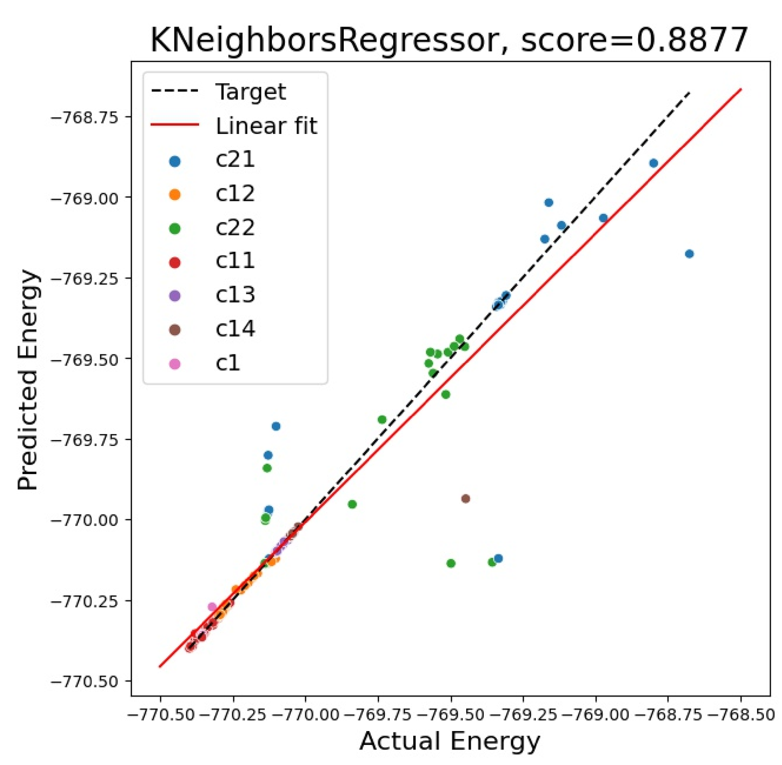


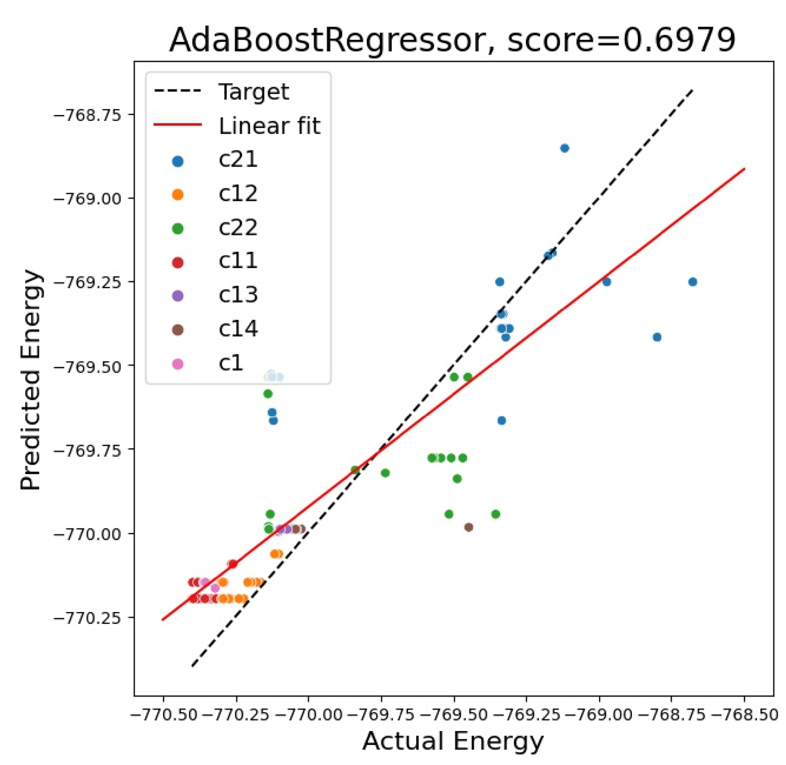
671 Features



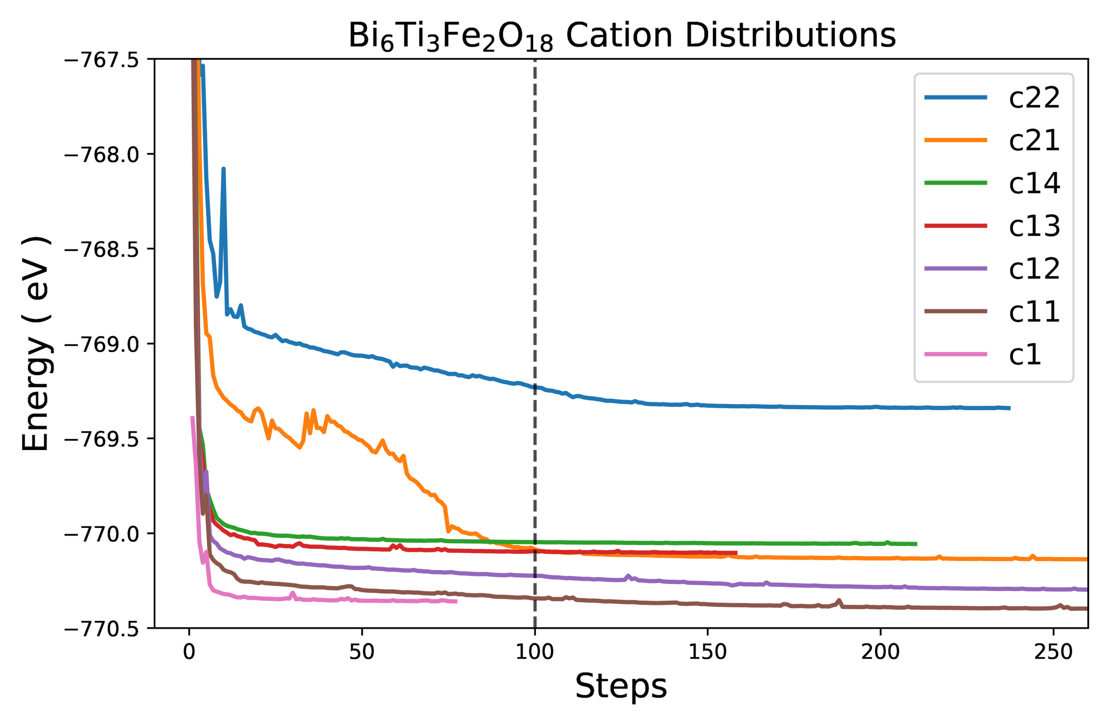
Method Performance:

	Fitting time	r2_score_train	mae_train	rmse_train	r2_score_valid	mae_valid	rmse_valid
DummyRegressor	0.000587	0.000000	0.305342	0.435107	-0.000269	0.311976	0.398841
LinearSVR	0.033041	0.237574	0.203376	0.379922	0.402890	0.171488	0.308154
AdaBoostRegressor	2.627393	0.741273	0.161564	0.221318	0.697913	0.160574	0.219183
Ridge	0.061749	0.652825	0.119998	0.256372	0.744767	0.110917	0.201470
ExtraTreesRegressor	13.343899	1.000000	0.000000	0.000001	0.806476	0.049202	0.175432
SVR	0.120129	0.697135	0.121641	0.239453	0.822437	0.111931	0.168042
RandomForestRegressor	38.600513	0.955257	0.026171	0.092036	0.850271	0.052410	0.154310
GradientBoostingRegressor	18.912478	0.959104	0.038899	0.087991	0.869910	0.059647	0.143835
KNeighborsRegressor	0.001020	0.788402	0.055962	0.200148	0.887726	0.042776	0.133623





Energy of Each Configuration:



Conclusions

- **Features related to Bismuth and Oxygen became redundant in our machine learning approach.**
- **Gradient Boosting Regression showed the best performance in all different scenarios.**

References

[1] Wang, Anthony Yu-Tung et al., **Machine Learning for Materials Scientists: An Introductory Guide toward Best Practices.** Chemistry of Materials, 2020.
[2] Andriy Burkov, **The Hundred-Page Machine Learning Book**, 2019.

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